LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Unit Test: Metals and Non-metals

Chapter / Topic: Metals and Non-metals (topic-based)

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SECTION A

Q1. Which metal is so soft that it can be cut with a knife? **[1]**

Answer: (d) Sodium

Q2. A solution of which non-metal is applied on wounds as an antiseptic? **[1]**

Answer: (c) Iodine

Q3. Which non-metal do we take in when we breathe? **[1]**

Answer: (a) Oxygen

Q4. Coating iron with a layer of zinc is called: **[1]**

Answer: (b) galvanisation

Q5. The green coating on copper vessels left in moist air is: **[1]**

Answer: (d) a mixture of copper hydroxide and copper carbonate

Q6. Which gas is produced when metals react with dilute acids? **[1]**

Answer: (c) Hydrogen

SECTION B

Q7. Why are electric wires made of copper, but covered with plastic? **[2]**

Answer: Copper is a good conductor of electricity (and ductile). Plastic is an insulator, so the covering protects us from electric shocks.

- Copper conducts: 1
- Plastic insulates: 1

Q8. Why should lemon pickle not be stored in aluminium or iron containers? **[2]**

Answer: Lemon pickle contains acids, which react with metals. This spoils the pickle (harmful substances may form) and damages the container.

- Acids react with metals: 1
- Spoils food / damages container: 1

Q9. An iron nail is placed in copper sulphate solution. What do you observe, and what does it show? **[2]**

Answer: The blue colour of the solution fades and turns greenish, and a reddish-brown coating of copper forms on the nail. This shows that iron is more reactive than copper: it displaces copper.

- Observation: 1
- Iron more reactive: 1

Q10. Give two uses each of metals and non-metals. **[2]**

Answer: Metals (any two): machinery, cooking utensils, electric wires, jewellery, automobiles. Non-metals (any two): oxygen for breathing; nitrogen and phosphorus in fertilisers; chlorine to purify water; iodine as an antiseptic.

- Two uses each: 1 + 1

SECTION C

Q11. Write word equations for what happens when: (i) sodium is put in water, (ii) a magnesium ribbon burns in air, (iii) sulphur burns in air. **[3]**

Answer: (i) Sodium + water → sodium hydroxide + hydrogen (plus a lot of heat). (ii) Magnesium + oxygen → magnesium oxide. (iii) Sulphur + oxygen → sulphur dioxide.

- Each: 1

Q12. The ash of a burnt magnesium ribbon is dissolved in water, and so is the gas from burning sulphur. Each solution is tested with litmus. What do you… **[3]**

Answer: The magnesium oxide solution turns red litmus blue, so it is basic. The sulphur dioxide solution turns blue litmus red, so it is acidic. Conclusion: metals form basic oxides and non-metals form acidic oxides.

- MgO basic: 1
- SO_2 acidic: 1
- Conclusion: 1

SECTION D

Q13. Answer the following: **[5]**

(i) A reaction in which a more reactive metal pushes out (displaces) a less reactive metal from its compound in solution. Examples: zinc + copper sulphate → zinc sulphate + copper; iron + copper sulphate → iron sulphate + copper. [3]

- Definition: 1
- Each example: 1 + 1

(ii) (i) No. Iron is less reactive than zinc, so it cannot displace zinc. (ii) Yes. Zinc is more reactive than iron: zinc + iron sulphate → zinc sulphate + iron. [2]

- Each with reason: 1